charged candidate $h^{\pm}$

$h^{\pm}$

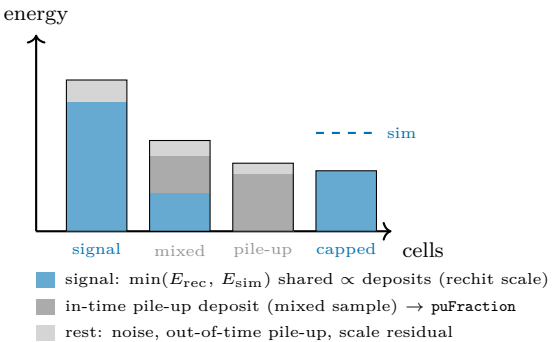
`track`
$\downarrow \chi^2$ `association`
`TrackingParticle`
$\rightarrow$ `SimTrack`

hard-scatter particle $\Rightarrow$ **100%** of the candidate energy is signal pile-up particle (mixed sample) $\Rightarrow$ 100% pile-up no match $\Rightarrow$ unattributed (fake, unlinked)

one object, one momentum: no hits needed

neutral candidate γ / h^0

h^0 PF cluster: cells $\times$ fractions



sum over the cells, rescaled to the candidate energy $\Rightarrow$ `signalFraction`, `puFraction` of the candidate

why the cell: the finest unit both sides share. The rechit is one number per cell (the digitiser already summed every Geant4 step), the sim hits say who deposited how much in it; anything coarser needs a shower-shape assumption, anything finer has no reconstructed counterpart

the jet

$$\vec{p}_T^{\,\rm sig} = \sum_{c \,\in\, \rm jet} w_c \, f_c^{\rm sig} \, \vec{p}_{T,c}$$

w_c: clustering weight (Puppi), $f_c^{\rm sig}$: `signalFraction`

`signalPtRatio` $= |\vec{p}_T^{\,\rm sig}|/p_T^{\rm GEN}$

`ptRatio` $-$ `signalPtRatio` $=$ the pile-up in this jet

shared-energy response: keep only the shares whose particle descends from the matched GenJet's constituents $\Rightarrow$ the gen jet's own energy, nothing else

signal = any particle of the hard-scatter event (underlying event and other jets included); nothing is subtracted, pile-up is simply never counted. The matching uses the signal shares only, so a pile-up dominated jet is still paired with the gen jet it contains